Instituto Politécnico Nacional
Centro de Investigación en Computación

Mathematics for Computer Science

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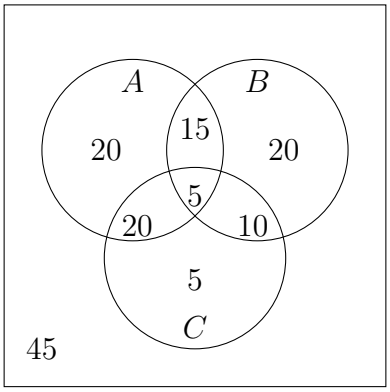
Homework 1

In a survey of 140 people, it was found that:
60 like Football, 50 like Basketball and 40 like Tennis,
20 like both Football and Basketball,
25 like both Football and Tennis,
15 like both Basketball and Tennis,
5 like all three sports.
Find the number of people who:

1. only like Football.
2. like Football or Tennis.
3. like at least one of these three sports.
4. do not like any of these three sports.

Solution:
Let:
A be the set of people that like Football
B be the set of people that like Basketball
C be the set of people that like Tennis

Then, drawing the Venn diagram, we get:



Thus we can say:

1. 20 people only like Football ($A \cap B^c \cap C^c$).
2. 75 people like Football or Tennis ($A \cup C$).
3. 95 people like at least one of these three sports ($A \cup B \cup C$).
4. 45 people don't like any of these three sports $(A \cup B \cup C)^c$.